18.11 Three Classical Inequalities

§10 built the basic theory of convex functions. This section spends it on three classical inequalities, each obtained from the one before. Jensen's inequality for the logarithm is the weighted arithmetic–geometric mean inequality; with two weights it is Young's inequality; and Young's inequality, applied coordinate by coordinate, is Hölder's. Hölder's inequality then settles two questions the book left open: the triangle inequality for the p-norm, which Chapter 16 §01 recorded without proof, and the dual of the p-norm, which §04 computed only for $p = 1, 2, \infty$.

Throughout, the field is $\mathbb{R}$, and "increasing" and "strictly increasing" are meant as in §10.

18.11.1 The logarithm and real powers

The first inequality needs a logarithm, and the book has none. It has the exponential: by fact (A1) of Chapter 10 §09, for real s the series $\sum_i s^i/i!$ converges to a real number e^s, with $e^{s+u} = e^s e^u$, $e^0 = 1$, and $e^s \geq 1$ for $s \geq 0$; and $\frac{d}{dt}e^t = e^t$ is the 1×1 case of Theorem 10.9.4 (b). We construct the logarithm as its inverse.

Lemma 18.11.1. Exponential and Logarithm

(a) $e^s > 0$ for every real s, and $s \mapsto e^s$ is strictly increasing and strictly convex on $\mathbb{R}$.

(b) For every real $y > 0$ there is exactly one real s with $e^s = y$; write $s = \log y$.

(c) $\log\colon (0, \infty) \to \mathbb{R}$ is strictly increasing and **strictly concave**, with $\log(xy) = \log x + \log y$ and $\log 1 = 0$.

Idea. The exponential is positive and is its own derivative, so it is strictly increasing with a strictly increasing derivative, hence strictly convex. The intermediate value theorem makes it take every positive value, so it has an inverse. Applying that increasing inverse to the convexity inequality of exp turns it into concavity of log.

Proof.

(a) If $s \geq 0$ then $e^s \geq 1 > 0$. If $s < 0$ then $e^{-s} \geq 1$, and $e^s e^{-s} = e^0 = 1$, so $e^s = 1/e^{-s} > 0$. The derivative e^s is therefore positive everywhere, and by the linearity of differentiation, fact (A3) of Chapter 10 §09, $-\exp$ has the negative derivative $-\exp$. By fact (A6) of Chapter 16's introduction, $-\exp$ is strictly decreasing, so exp is strictly increasing. So $g = \exp$ has a strictly increasing derivative, and Lemma 18.10.5 makes it strictly convex.

(b) Uniqueness holds because exp is strictly increasing. For existence, note first that exp is continuous: $e^{t+h} - e^t = e^t\, h \cdot (e^h - 1)/h$, and $(e^h - 1)/h \to 1$ is the derivative at 0, so the right side tends to 0. Let $y \geq 1$. For $s \geq 0$ every partial sum of the series with at least two terms is at least $1 + s$, so $e^s \geq 1 + s$, since a nonstrict inequality survives a limit. Hence $e^y \geq 1 + y > y$, while $e^0 = 1 \leq y$. The intermediate value theorem, fact (A5) of Chapter 16's introduction, gives $s \in [0, y]$ with $e^s = y$. If $0 < y < 1$, apply this to $1/y > 1$ to get $e^{s'} = 1/y$; then $e^{-s'} = y$.

(c) If $0 < x < y$ and $\log x \geq \log y$, then $x = e^{\log x} \geq e^{\log y} = y$, since exp is increasing, which is false. So log is strictly increasing. Next, $e^{\log x + \log y} = e^{\log x} e^{\log y} = xy$, so $\log(xy) = \log x + \log y$ by uniqueness in (b), and $e^0 = 1$ gives $\log 1 = 0$. For concavity,

let $x, y > 0$ and $0 \le t \le 1$, and put $a = \log x$, $b = \log y$. By (a),

$$e^{ta+(1-t)b} \le te^a + (1-t)e^b = tx + (1-t)y.$$

Applying the increasing function log to both sides gives $t\log x + (1-t)\log y \le \log(tx + (1-t)y)$. If $x \ne y$ and $0 < t < 1$, then $a \ne b$, the first inequality is strict by strict convexity, and log keeps it strict. So $-\log$ is strictly convex.

■

Real powers. For $t > 0$ and real r put $t^r = e^{r\log t}$, and $0^r = 0$ for $r > 0$. For a positive integer r this is the usual power, since $e^{r\log t} = (e^{\log t})^r = t^r$ by $e^{s+u} = e^s e^u$. For $t, a, b > 0$ and real r, u, the rules $e^{s+u} = e^s e^u$ and $\log(ab) = \log a + \log b$, with $\log(t^r) = r\log t$ from the uniqueness in (b), give

$$t^r t^u = t^{r+u}, \qquad (t^r)^u = t^{ru}, \qquad (ab)^r = a^r b^r, \qquad (a/b)^r = a^r/b^r.$$

The first three extend to $t, a, b \ge 0$, and the last to $a \ge 0$, when the exponents r and u are positive: then a zero base makes both sides 0. They do not extend further, since 0^r is not defined for $r \le 0$. Finally, for $r > 0$ the function $t \mapsto t^r$ is strictly increasing on $[0, \infty)$: on $(0, \infty)$ it is exp applied to the strictly increasing $r\log t$, and $0^r = 0 < t^r$ for $t > 0$. This is the meaning of $|x_i|^p$ and of the exponent $1/p$ in the p-norm $\|\mathbf{x}\|_p = (\sum_i |x_i|^p)^{1/p}$ of Chapter 16 §01.

18.11.2 Arithmetic and geometric means

Chapter 17 §08 proved Lemma 17.8.6, the arithmetic–geometric mean inequality, by pure algebra: an induction that merges a number above the mean with one below it. Concavity of the logarithm gives a second proof, and a stronger statement, with arbitrary weights.

Theorem 18.11.2. Weighted Arithmetic–Geometric Mean Inequality

Let $t_1, \dots, t_m \ge 0$, and let $w_1, \dots, w_m > 0$ with $w_1 + \cdots + w_m = 1$. Then

$$t_1^{w_1} t_2^{w_2} \cdots t_m^{w_m} \;\le\; w_1 t_1 + w_2 t_2 + \cdots + w_m t_m,$$

with equality if and only if $t_1 = t_2 = \cdots = t_m$.

Idea. Take logarithms. The product becomes the weighted average $\sum_i w_i \log t_i$, and the inequality becomes Jensen's inequality for the concave log. Then exponentiate back.

Proof. If some $t_j = 0$, the left side is 0, since $0^{w_j} = 0$, and the right side is ≥ 0. Equality then means $\sum_i w_i t_i = 0$, which with every $w_i > 0$ and $t_i \ge 0$ forces every $t_i = 0$; conversely, if every $t_i = 0$, both sides are 0.

So let every $t_i > 0$. By Lemma 18.11.1 (c), $-\log$ is strictly convex on $(0, \infty)$, so Theorem 18.10.2 gives

$$\sum_{i=1}^{m} w_i \log t_i \;\le\; \log\Big(\sum_{i=1}^{m} w_i t_i\Big),$$

with equality only when all t_i are equal. Apply the strictly increasing function exp. The right side becomes $\sum_i w_i t_i$, and the left side becomes $\prod_i e^{w_i \log t_i} = \prod_i t_i^{w_i}$, by $e^{s+u} = e^s e^u$

and the definition of real powers. A strictly increasing function preserves both the inequality and its strictness, so equality holds exactly when all t_i are equal.

■

The case of equal weights is Lemma 17.8.6. With $w_i = 1/n$, the rules for powers turn the theorem into $(t_1 \cdots t_n)^{1/n} \le \frac{1}{n}\sum_i t_i$. Since $u \mapsto u^n$ is strictly increasing on $[0,\infty)$, raising both sides to the n-th power gives $t_1 \cdots t_n \le (\frac{1}{n}\sum_i t_i)^n$, with the same equality case. That is Lemma 17.8.6, now with arbitrary weights and seen as Jensen's inequality for the logarithm.

With two weights, the theorem becomes an inequality about products that is exactly what Hölder's inequality needs.

Lemma 18.11.3. Young's Inequality

Let $p, q > 1$ with $\frac{1}{p} + \frac{1}{q} = 1$. Then for all real $a, b \ge 0$,

$$ab \;\le\; \frac{a^p}{p} + \frac{b^q}{q},$$

with equality if and only if $a^p = b^q$.

Proof. Apply Theorem 18.11.2 with $m = 2$, $t_1 = a^p$, $t_2 = b^q$, $w_1 = 1/p$, and $w_2 = 1/q$. The weights are positive and add up to 1, and $(a^p)^{1/p} = a$, $(b^q)^{1/q} = b$ by the rules for powers. So the left side of the theorem is ab and the right side is $a^p/p + b^q/q$, with equality exactly when $a^p = b^q$.

■

18.11.3 Hölder's inequality and the dual p-norms

Two exponents $p, q \in (1,\infty)$ with $\frac{1}{p} + \frac{1}{q} = 1$ are called **conjugate**; equivalently $q = p/(p-1)$. The pair $p = q = 2$ is conjugate to itself. Hölder's inequality is the Cauchy–Schwarz inequality with the two factors measured in two different norms.

Theorem 18.11.4. Hölder's Inequality

Let $p, q \in (1,\infty)$ be conjugate exponents. For all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$,

$$|\langle \mathbf{x}, \mathbf{y} \rangle| \;\le\; \sum_{i=1}^{n} |x_i y_i| \;\le\; \|\mathbf{x}\|_p \|\mathbf{y}\|_q.$$

Idea. Both sides of the second inequality scale the same way when $\mathbf{x}$ or $\mathbf{y}$ is multiplied by a positive number. So normalize to $\|\mathbf{x}\|_p = \|\mathbf{y}\|_q = 1$. Then apply Young's inequality coordinate by coordinate and add: the right-hand sides add up to $\frac{1}{p} + \frac{1}{q} = 1$.

Proof. The first inequality is the triangle inequality for numbers (Theorem 0.6.4) applied to $\sum_i x_i y_i$. For the second, if $\mathbf{x} = \mathbf{0}$ or $\mathbf{y} = \mathbf{0}$ both sides are 0. Otherwise $\|\mathbf{x}\|_p > 0$ and $\|\mathbf{y}\|_q > 0$; put $a_i = |x_i|/\|\mathbf{x}\|_p$ and $b_i = |y_i|/\|\mathbf{y}\|_q$. Then $\sum_i a_i^p = \sum_i |x_i|^p/\|\mathbf{x}\|_p^p = 1$, because $\|\mathbf{x}\|_p^p = \sum_i |x_i|^p$ by the rules for powers, and likewise $\sum_i b_i^q = 1$. By Lemma 18.11.3 for each i, and adding,

$$\sum_{i=1}^{n} a_i b_i \;\le\; \frac{1}{p}\sum_{i=1}^{n} a_i^p + \frac{1}{q}\sum_{i=1}^{n} b_i^q = \frac{1}{p} + \frac{1}{q} = 1.$$

Multiplying by $\|\mathbf{x}\|_p \|\mathbf{y}\|_q$ gives $\sum_i |x_i y_i| \le \|\mathbf{x}\|_p \|\mathbf{y}\|_q$, as claimed.

■

For $p = q = 2$ this is Theorem 11.1.5 for the standard inner product on $\mathbb{R}^n$, reached without the inner product's geometry. The limiting pair $p = 1$, $q = \infty$ holds too, and more simply: $\sum_i |x_i y_i| \le \sum_i |x_i| \|\mathbf{y}\|_\infty$.

Chapter 16 §01 recorded the p-norm for every $p \ge 1$, but verified only $p = 1, 2, \infty$, because the triangle inequality for general p had not been proved. Hölder's inequality proves it.

Corollary 18.11.5. Minkowski's Inequality

Let $1 < p < \infty$. For all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, $\|\mathbf{x} + \mathbf{y}\|_p \le \|\mathbf{x}\|_p + \|\mathbf{y}\|_p$. Consequently $\|\cdot\|_p$ is a norm on $\mathbb{R}^n$.

Idea. Split $|z_i|^p = |z_i| \cdot |z_i|^{p-1}$ and use the triangle inequality on the first factor. That leaves two sums, and Hölder's inequality bounds each, with the same second factor $(|z_i|^{p-1})_i$. Dividing by that common factor's norm leaves the triangle inequality.

Proof. Let $q = p/(p-1)$ be the conjugate exponent, and put $\mathbf{z} = \mathbf{x} + \mathbf{y}$. If $\mathbf{z} = \mathbf{0}$ there is nothing to prove. Otherwise, by the triangle inequality for numbers in each coordinate (Theorem 0.6.4),

$$\begin{aligned}\sum_i |z_i|^p &= \sum_i |z_i|\,|z_i|^{p-1} \\ &\le \sum_i |x_i|\,|z_i|^{p-1} + \sum_i |y_i|\,|z_i|^{p-1}.\end{aligned}$$

Apply Theorem 18.11.4 to each sum, with the vector $\mathbf{w} = (|z_1|^{p-1}, \dots, |z_n|^{p-1})$ as the second factor. Since $(p-1)q = p$,

$$\|\mathbf{w}\|_q = \Big(\sum_i |z_i|^p\Big)^{1/q} = \|\mathbf{z}\|_p^{p/q}.$$

So $\|\mathbf{z}\|_p^p \le (\|\mathbf{x}\|_p + \|\mathbf{y}\|_p)\|\mathbf{z}\|_p^{p/q}$. Divide by $\|\mathbf{z}\|_p^{p/q} > 0$, and use $p - p/q = p(1 - 1/q) = 1$, to get $\|\mathbf{z}\|_p \le \|\mathbf{x}\|_p + \|\mathbf{y}\|_p$. This is (N3) of Definition 16.1.1. (N1) holds because $\|\mathbf{x}\|_p = 0$ forces every $|x_i|^p = 0$, and (N2) because $\big(\sum_i |c|^p |x_i|^p\big)^{1/p} = |c| \|\mathbf{x}\|_p$ by the rules for powers.

■

The proof uses the coordinates only through their moduli: the triangle inequality $|x_i + y_i| \le |x_i| + |y_i|$, which Theorem 0.6.4 proves for complex numbers as well, and Hölder's inequality for the real vectors $(|x_i|)_i$, $(|y_i|)_i$, and $\mathbf{w}$, whose p-norms are those of $\mathbf{x}$ and $\mathbf{y}$. So it works verbatim on $\mathbb{C}^n$, and $\|\cdot\|_p$ is a norm on F^n for $F = \mathbb{R}$ and for $F = \mathbb{C}$, as Chapter 16 §01 recorded.

Now the dual norm. Recall from Definition 18.4.4 that the dual norm of $\|\cdot\|$ at $\mathbf{y}$ is the largest value of $\langle \mathbf{x}, \mathbf{y} \rangle$ over the unit ball $\|\mathbf{x}\| \le 1$. §04 computed it for $p = 1, 2, \infty$: the 1-norm and the ∞-norm are dual to each other, and the 2-norm is dual to itself. Hölder's inequality supplies an upper bound for every other p, and a matching vector shows it is attained.

Theorem 18.11.6. The Dual of the p-Norm

Let $p, q \in (1, \infty)$ be conjugate exponents. On $\mathbb{R}^n$ with the standard inner product, the dual norm of $\|\cdot\|_p$ is $\|\cdot\|_q$:

$$\max\{\langle \mathbf{x}, \mathbf{y} \rangle : \|\mathbf{x}\|_p \le 1\} = \|\mathbf{y}\|_q \qquad (\mathbf{y} \in \mathbb{R}^n).$$

For $\mathbf{y} \neq \mathbf{0}$ the maximum is attained at $\mathbf{x}$ with $x_i = \sigma_i |y_i|^{q-1} / \|\mathbf{y}\|_q^{q-1}$, where $\sigma_i = 1$ if $y_i \ge 0$ and $\sigma_i = -1$ otherwise.

Idea. Hölder's inequality bounds $\langle \mathbf{x}, \mathbf{y} \rangle$ by $\|\mathbf{y}\|_q$ on the whole unit ball. To show the bound is attained, give x_i the sign of y_i and make $|x_i|^p$ proportional to $|y_i|^q$, which is when Young's inequality (Lemma 18.11.3) is an equality in each coordinate. That is the stated witness, and a direct computation checks it.

Proof. If $\|\mathbf{x}\|_p \le 1$, then $\langle \mathbf{x}, \mathbf{y} \rangle \le \|\mathbf{x}\|_p \|\mathbf{y}\|_q \le \|\mathbf{y}\|_q$ by Theorem 18.11.4. For $\mathbf{y} = \mathbf{0}$ the bound 0 is attained at $\mathbf{x} = \mathbf{0}$. For $\mathbf{y} \neq \mathbf{0}$, take $\mathbf{x}$ as in the statement. Since $(q-1)p = q$,

$$\sum_i |x_i|^p = \frac{\sum_i |y_i|^q}{\|\mathbf{y}\|_q^q} = 1,$$

so $\|\mathbf{x}\|_p = 1$; and since $\sigma_i y_i = |y_i|$,

$$\langle \mathbf{x}, \mathbf{y} \rangle = \frac{\sum_i |y_i|^q}{\|\mathbf{y}\|_q^{q-1}} = \frac{\|\mathbf{y}\|_q^q}{\|\mathbf{y}\|_q^{q-1}} = \|\mathbf{y}\|_q.$$

So the bound is attained, and the largest value is $\|\mathbf{y}\|_q$.

■

Example. A dual norm by hand

Let $\mathbf{y} = (3, -4, 5) \in \mathbb{R}^3$, and let $p = \frac{3}{2}$ and $q = 3$. Compute the dual norm of $\mathbf{y}$ with respect to $\|\cdot\|_{3/2}$, find a vector $\mathbf{x}$ with $\|\mathbf{x}\|_{3/2} = 1$ at which it is attained, and compare with the value at the unit vector $\mathbf{e}_3$.

Solution. The exponents are conjugate, since $\frac{1}{p} + \frac{1}{q} = \frac{2}{3} + \frac{1}{3} = 1$. Next, $|3|^3 + |-4|^3 + |5|^3 = 27 + 64 + 125 = 216 = 6^3$, so $\|\mathbf{y}\|_3 = 216^{1/3} = 6$, using $(6^3)^{1/3} = 6$. By Theorem 18.11.6 the

dual norm is 6.

For the maximizer, $q-1=2$, so $x_i=\sigma_i|y_i|^2/6^2$ with σ_i the sign of y_i:

$$\mathbf{x}=(\tfrac{9}{36},-\tfrac{16}{36},\tfrac{25}{36})=(\tfrac{1}{4},-\tfrac{4}{9},\tfrac{25}{36}).$$

Check: $|x_i|^{3/2}=(|y_i|^2/36)^{3/2}=|y_i|^3/216$, giving $\frac{1}{8}+\frac{8}{27}+\frac{125}{216}=\frac{27+64+125}{216}=1$. So $\|\mathbf{x}\|_{3/2}=1$. And $\langle\mathbf{x},\mathbf{y}\rangle=\frac{3}{4}+\frac{16}{9}+\frac{125}{36}=\frac{27+64+125}{36}=6$.

The unit vector $\mathbf{e}_3$ has $\|\mathbf{e}_3\|_{3/2}=1$ and gives $\langle\mathbf{e}_3,\mathbf{y}\rangle=5<6$. Putting all the weight on the largest coordinate of $\mathbf{y}$ is not optimal; the maximizer spreads it in proportion to $|y_i|^2$.

Exchanging the roles of p and q shows that the dual of $\|\cdot\|_q$ is $\|\cdot\|_p$, just as Theorem 18.4.8 requires. Together with §04, the dual norm of $\|\cdot\|_p$ is now known for every $p\in[1,\infty]$: it is $\|\cdot\|_q$ with $\frac{1}{p}+\frac{1}{q}=1$, reading $\frac{1}{\infty}$ as 0. This completes §04's computation. Chapter 16 measured vectors and matrices but never a linear functional; §04's dual norm filled that gap, and now it is computed for every p-norm.

Look back at the chain. One convexity fact, that $-\log$ is strictly convex, went through Jensen's inequality to the weighted arithmetic–geometric mean inequality, through two weights to Young's inequality, and through normalization to Hölder's. Hölder's inequality then did two jobs that earlier chapters could not: it proved that every $\|\cdot\|_p$ is a norm, so that its unit ball is one of the convex sets this chapter began with, and it computed the dual of every such norm.

18.11.4 Summary and transfer

The chapter began with a word the book had been using without studying it. Convexity turned up in the positive semidefinite cone of Chapter 13, in the unit balls of Chapter 16, and in the Ky Fan sums of Chapter 17. The chapter ends with convexity as a subject: convex sets, the hyperplanes that separate them, the duality between a cone and its dual, extreme points and polytopes, and convex functions. Six moves did the work, and each transfers beyond this chapter.

- **Nearest point, then the normal direction.** Project onto a closed convex set; the vector from the projection to the point is the normal of a separating hyperplane (§03, Separating Hyperplanes). This is Chapter 11's best approximation, with an inequality where the equation was. *Transfer:* to prove that a point lies outside a closed convex set, and to certify it, look for the nearest point.
- **Count, then drop a point.** Too many points in n dimensions are affinely dependent, and moving the weights along the dependence removes one of them without leaving the hull (§02, Carathéodory's Theorem). Among too many generators of a cone, a linear dependence does the same (§05, Cones and Farkas's Lemma). *Transfer:* when a representation uses more pieces than the dimension allows, the surplus is a dependence, and a dependence can be spent.
- **Dualize.** A closed convex cone is determined by its dual cone, and every statement about the one has a statement about the other. Farkas's lemma is this move made into a theorem, linear programming duality (§06, Linear Programming Duality) is Farkas applied to one combined system, and the minimax theorem (§07, Matrix Games and the Minimax Theorem) is a duality statement of the same kind. *Transfer:* when a feasibility question will not yield, ask for the certificate that it fails.
- **Lift, then eliminate.** A hull of finitely many points is the projection of a polyhedron in a bigger space, with the weights as extra coordinates, and Fourier–Motzkin elimination

removes those coordinates one at a time while keeping a system of inequalities (§09, Polytopes, Theorem 18.9.5). *Transfer:* to describe the image of a set given by inequalities, add the hidden variables as coordinates and eliminate them.

- **A supremum of linear functions is convex.** Support functions, dual norms, λ_1, and the Ky Fan sums are convex for this one reason (§04, Gauges, Support Functions, and Dual Norms, and §10, Convex Functions, Proposition 18.10.3). Its companion: a linear function on a compact convex set peaks at an extreme point (§08, Extreme Points). *Transfer:* to prove a function convex, write it as a maximum of affine functions.
- **Compactness turns a supremum into a maximum.** Once a hull, a ball, or a set of weights is known to be compact, facts (A3) and (A4) of Chapter 16's introduction produce a nearest point, a maximizer, or an extreme point that can be named and used. *Transfer:* before optimizing, check that the set is compact; afterwards, feed the maximizer back into the argument.

§§10–11 added two moves of their own. **Second differences**: on a segment, convexity is read off from $f(\mathbf{z}+s\mathbf{v})+f(\mathbf{z}-s\mathbf{v})-2f(\mathbf{z})$ (§10, Theorem 18.10.6). **Normalize, then use a two-point inequality**: Hölder's inequality is Young's inequality after scaling both vectors to norm 1 (§11, Theorem 18.11.4). *Transfer:* if multiplying any one variable by $c>0$ multiplies both sides by the same power of c, check the cases with a zero variable directly, and then assume every variable has norm 1.

The chapter also paid promises made in earlier chapters.

The promise	Made in	Paid by
"Chapter 18 studies convex functions in general, and the observation that a pointwise supremum of linear functions is convex is one of its basic facts"	Chapter 17 §06	§10 (Convex Functions), Proposition 18.10.3, with λ_1 and the Ky Fan sums as its instances (Corollary 18.10.4)
Nonnegativity and capacities turn traffic flow "into one about inequalities, which is the subject of linear programming"	Chapter 3 §07	§06 (Linear Programming Duality)
A way to measure a linear functional: Chapter 16 measured vectors and matrices, but never a functional	Chapter 16	§04 (Gauges, Support Functions, and Dual Norms): the definition, the double dual, and $p=1,2,\infty$; §11 (Three Classical Inequalities): every $1<p<\infty$, Theorem 18.11.6
The triangle inequality for $\|\cdot\|_p$, recorded without proof, "which Chapter 18 proves from Hölder's inequality"	Chapter 16 §01	§11 (Three Classical Inequalities), Corollary 18.11.5

Some things the chapter did not do. It stated Birkhoff's theorem, that the extreme points of the doubly stochastic matrices are exactly the permutation matrices, and left its proof to Chapter 19. The convexity of $\mathbf{A}\mapsto \operatorname{tr} f(\mathbf{A})$ for convex f, and the consequences of a majorization for

every convex function, belong to Chapter 21. Convex optimization beyond linear programs, where a duality gap can occur, lies outside this chapter.

18.11.5 Exercises

18.11.5.1 A. Check your understanding

Exercise. A1

(a) State Young's inequality, with its hypotheses and its equality case.

(b) Decide whether the following is correct, and justify your answer: the exponent conjugate to $p = 4$ is $q = \frac{4}{3}$.

(c) Decide whether the following is correct, and justify your answer: $(t^r)^u = t^{ru}$ for every real $t \geq 0$ and all real r, u.

(d) Decide whether the following is correct, and justify your answer: if p, q are conjugate and $\langle \mathbf{x}, \mathbf{y} \rangle = \|\mathbf{x}\|_p \|\mathbf{y}\|_q$ for $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, then one of $\mathbf{x}, \mathbf{y}$ is a scalar multiple of the other.

(e) Name the convex function to which the proof of Theorem 18.11.2 applies Jensen's inequality, and say why the equality case needs it to be **strictly** convex.

Solution.

(a) Let $p, q > 1$ with $\frac{1}{p} + \frac{1}{q} = 1$. Then $ab \leq a^p/p + b^q/q$ for all real $a, b \geq 0$, with equality if and only if $a^p = b^q$ (Lemma 18.11.3).

(b) Correct. $\frac{1}{4} + \frac{3}{4} = 1$; equivalently $q = p/(p-1) = 4/3$.

(c) Incorrect. Take $t = 0$, $r = 1$, and $u = -1$. Then $t^r = 0$, and 0^{-1} is not defined, since 0^r is defined only for $r > 0$; neither side has a value. The rule holds for $t > 0$ and all real r, u, and for $t = 0$ when $r, u > 0$.

(d) Incorrect. Take $p = 3$, $q = \frac{3}{2}$, $\mathbf{x} = (1, 2)$, and $\mathbf{y} = (1, 4)$. Then $\langle \mathbf{x}, \mathbf{y} \rangle = 1 + 8 = 9$, $\|\mathbf{x}\|_3 = (1 + 8)^{1/3} = 9^{1/3}$, and $\|\mathbf{y}\|_{3/2} = (1 + 4^{3/2})^{2/3} = 9^{2/3}$, since $4^{3/2} = 8$; so $\|\mathbf{x}\|_3 \|\mathbf{y}\|_{3/2} = 9^{1/3+2/3} = 9$. But neither vector is a multiple of the other, since $1 \cdot 4 - 2 \cdot 1 \neq 0$. Equality asks for $|x_i|^p$ to be proportional to $|y_i|^q$, not $\mathbf{x}$ to $\mathbf{y}$ (exercise C2).

(e) It is $-\log$ on $(0, \infty)$, which is strictly convex by Lemma 18.11.1 (c). Theorem 18.10.2 proves its equality clause only for strictly convex functions, and the hypothesis is needed: an affine function makes Jensen's inequality an equality for every choice of points.

18.11.5.2 B. Practice

Exercise. B1: A dual norm by hand

Let $\mathbf{y} = (1, -6, 8) \in \mathbb{R}^3$, and let $p = \frac{3}{2}$ and $q = 3$.

(a) Check that p and q are conjugate, and compute $\|\mathbf{y}\|_3$.

(b) Find $\mathbf{x} \in \mathbb{R}^3$ with $\|\mathbf{x}\|_{3/2} = 1$ and $\langle \mathbf{x}, \mathbf{y} \rangle = \|\mathbf{y}\|_3$.

(c) Hence determine the dual norm of $\mathbf{y}$ with respect to $\|\cdot\|_{3/2}$, and compare it with the value $\langle \mathbf{e}_3, \mathbf{y} \rangle$ at the unit vector $\mathbf{e}_3$.

Solution.

(a) $\frac{1}{p}+\frac{1}{q}=\frac{2}{3}+\frac{1}{3}=1$. Next, $|1|^3+|-6|^3+|8|^3=1+216+512=729=9^3$, so $\|\mathbf{y}\|_3=729^{1/3}=9$, using $(9^3)^{1/3}=9$.

(b) Follow Theorem 18.11.6: $x_i=\sigma_i|y_i|^2/9^2$ with σ_i the sign of y_i, since $q-1=2$. So

$$\mathbf{x}=(\tfrac{1}{81},-\tfrac{36}{81},\tfrac{64}{81})=(\tfrac{1}{81},-\tfrac{4}{9},\tfrac{64}{81}).$$

Check: $|x_i|^{3/2}=(|y_i|^2/81)^{3/2}=|y_i|^3/729$, giving $\frac{1}{729}+\frac{216}{729}+\frac{512}{729}=1$. So $\|\mathbf{x}\|_{3/2}=1$. And $\langle\mathbf{x},\mathbf{y}\rangle=\frac{1}{81}+\frac{216}{81}+\frac{512}{81}=\frac{729}{81}=9$.

(c) By Theorem 18.11.6 the dual norm is $\|\mathbf{y}\|_3=9$, attained at the $\mathbf{x}$ of (b). The unit vector $\mathbf{e}_3$ has $\|\mathbf{e}_3\|_{3/2}=1$ and gives $\langle\mathbf{e}_3,\mathbf{y}\rangle=8<9$: even when one coordinate of $\mathbf{y}$ dominates, putting all the weight on it is not optimal.

Exercise. B2: Harmonic and arithmetic means

Let $x_1,\dots,x_n>0$. Prove that

$$\frac{n}{\frac{1}{x_1}+\cdots+\frac{1}{x_n}}\le\frac{x_1+\cdots+x_n}{n},$$

with equality if and only if $x_1=\cdots=x_n$.

Solution. By Exercise (B1: Which are convex?) (d), $g(t)=1/t$ is strictly convex on $(0,\infty)$. Let $m=\frac{1}{n}(x_1+\cdots+x_n)>0$. Theorem 18.10.2 with the weights $w_i=1/n>0$ gives

$$\frac{1}{m}=g\Big(\sum_{i=1}^n\frac{1}{n}x_i\Big)\le\sum_{i=1}^n\frac{1}{n}\,g(x_i)=\frac{1}{n}\Big(\frac{1}{x_1}+\cdots+\frac{1}{x_n}\Big),$$

with equality only when all x_i are equal. Both sides are positive, and taking reciprocals reverses the inequality: $m\ge n/(\sum_i 1/x_i)$, which is the claim. If all x_i equal c, both sides equal c. So equality holds exactly when the x_i are all equal.

18.11.5.3 C. Going deeper

Exercise. C1: The exponential above its tangent

(a) Deduce from Exercise (C2: Tangent lines) (a) that $e^y\ge 1+y$ for every real y.

(b) Deduce that $\log x\le x-1$ for every real $x>0$.

Solution.

(a) By Lemma 18.11.1 (a), the function exp is convex and differentiable on the open interval $\mathbb{R}$, with derivative exp. So Exercise (C2: Tangent lines) (a) applies, and at $x=0$ it gives $e^y\ge e^0+e^0(y-0)=1+y$. For $y\ge 0$ this was visible from the series, as in the proof of Lemma 18.11.1 (b); for $y<0$ it is new.

(b) Let $x>0$ and put $y=\log x$, so that $e^y=x$ by Lemma 18.11.1 (b). Part (a) gives

$x \geq 1 + \log x$, that is, $\log x \leq x - 1$.

Exercise. C2: When Hölder is an equality

Let $p, q \in (1, \infty)$ be conjugate exponents, and let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ be nonzero.

(a) Prove that $\sum_i |x_i y_i| = \|\mathbf{x}\|_p \|\mathbf{y}\|_q$ if and only if $|x_i|^p / \|\mathbf{x}\|_p^p = |y_i|^q / \|\mathbf{y}\|_q^q$ for every i.

(b) Deduce that the maximizer in Theorem 18.11.6 is unique: if $\mathbf{y} \neq \mathbf{0}$, $\|\mathbf{x}\|_p \leq 1$, and $\langle \mathbf{x}, \mathbf{y} \rangle = \|\mathbf{y}\|_q$, then $\mathbf{x}$ is the vector given there.

Hint: in (a), look at where the proof of Theorem 18.11.4 adds up the inequalities of Lemma 18.11.3.

Solution.

(a) Put $a_i = |x_i| / \|\mathbf{x}\|_p$ and $b_i = |y_i| / \|\mathbf{y}\|_q$, as in the proof of Theorem 18.11.4, so that $\sum_i a_i^p = \sum_i b_i^q = 1$. Dividing by $\|\mathbf{x}\|_p \|\mathbf{y}\|_q > 0$, the equality $\sum_i |x_i y_i| = \|\mathbf{x}\|_p \|\mathbf{y}\|_q$ says $\sum_i a_i b_i = 1 = \sum_i (a_i^p/p + b_i^q/q)$. By Lemma 18.11.3, $a_i b_i \leq a_i^p/p + b_i^q/q$ for each i. A sum of nonnegative differences $(a_i^p/p + b_i^q/q) - a_i b_i$ is 0 exactly when each difference is 0, so equality holds exactly when $a_i b_i = a_i^p/p + b_i^q/q$ for every i, that is, by the equality case of Lemma 18.11.3, when $a_i^p = b_i^q$ for every i. By the rule $(a/b)^p = a^p/b^p$ for real powers, $a_i^p = |x_i|^p / \|\mathbf{x}\|_p^p$ and $b_i^q = |y_i|^q / \|\mathbf{y}\|_q^q$.

(b) Let $\mathbf{y} \neq \mathbf{0}$, $\|\mathbf{x}\|_p \leq 1$, and $\langle \mathbf{x}, \mathbf{y} \rangle = \|\mathbf{y}\|_q$. By Theorem 18.11.4,

$$\|\mathbf{y}\|_q = \langle \mathbf{x}, \mathbf{y} \rangle \leq \sum_i |x_i y_i| \leq \|\mathbf{x}\|_p \|\mathbf{y}\|_q \leq \|\mathbf{y}\|_q,$$

so every inequality is an equality. First, $\|\mathbf{x}\|_p = 1$, since $\|\mathbf{y}\|_q > 0$; in particular $\mathbf{x} \neq \mathbf{0}$. Second, $\sum_i x_i y_i = \sum_i |x_i y_i|$, and since $x_i y_i \leq |x_i y_i|$ for each i, every $x_i y_i \geq 0$. Third, by (a), $|x_i|^p = |y_i|^q / \|\mathbf{y}\|_q^q$ for every i. Raising to the power $1/p$ and using $q/p = q - 1$, which follows from $\frac{1}{p} + \frac{1}{q} = 1$, gives $|x_i| = |y_i|^{q-1} / \|\mathbf{y}\|_q^{q-1}$. If $y_i = 0$, then $x_i = 0$, as in the formula. If $y_i \neq 0$, then $x_i \neq 0$ and $x_i y_i \geq 0$, so x_i has the sign σ_i of y_i. Hence $x_i = \sigma_i |y_i|^{q-1} / \|\mathbf{y}\|_q^{q-1}$ for every i, which is the vector of Theorem 18.11.6.